INCH-POUND
A-A-59108C
23 SEP 2010
SUPERSEDING
A-A-59108B
22 JAN 2009

COMMERCIAL ITEM DESCRIPTION

TRUCK, FIRE FIGHTING (AIRCRAFT RESCUE) (ARFF)

The General Services Administration has authorized the use of this commercial item description for all federal agencies.

1. SCOPE. This commercial item description (CID) covers two sizes of diesel engine driven ARFF vehicles. It includes a 1500 and 3000 gallon commercially available version of ARFF vehicle. The ARFF vehicle is intended to carry rescue and firefighting equipment for rescuing occupants and combating fires in aircraft.

2. CLASSIFICATION. The ARFF trucks covered by this CID are classified as follows:

Minimum Rated Water Capacity

1,500 Gallons
3,000 Gallons

3. SALIENT CHARACTERISTICS. The ARFF truck shall be in accordance with the applicable requirements of National Fire Protection Association (NFPA) 414, Current Edition at the time of procurement

3.1 Administration.

3.1.1 Manuals.

3.1.1.1 Technical manuals. The overall format for the manuals may be military specification, commercial, or a combination of both. Each technical manual shall have a title page. One set of engine and transmission parts, service and operator's manuals shall be packed with each vehicle. An additional two complete sets of engine and transmission parts, service and operator's manuals shall be submitted to the procuring activity for stock.

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Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any data that may improve this document should be sent to: WR-ALC 542 CSW/CBSSS, 295 Byron Street, Robins AFB, GA 31098-1611. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.daps.dla.mil/online .

FSC 4210

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The contractor will submit digitized technical manuals and printed paper copies. The contractor may recommend the delivery of files developed in a "native" format as the final delivery medium. Native format includes MS Word, Framemaker, Interleaf, et cetera. However, all recommendations for native formatted data or graphics shall be approved by the Technical Order Manager prior to any work being performed.

a. The contractor shall validate the technical manuals for accuracy prior to submission to the procuring activity for verification. The contractor shall submit one complete set to the procuring activity for verification 30 days before the first production inspection. Any changes or corrections noted by the procuring activity shall be corrected and updated pages or manuals shall be submitted to the procuring activity.

b. Once approved by the procuring activity and a Technical Order number is assigned, the contractor shall pack two complete sets of both hard copy and CDs with each vehicle and one complete set of both hard copy and CDs shall be submitted to the procuring activity for stock.

c. The contractor shall grant the United States Air Force a non-exclusive, non-assignable, royalty free U.S. Government (Government) Purpose License, to scan into CENTRA (the USAF repository) and to reproduce and distribute (either electronically or via hard medium) copies or facsimiles of manuals produced and distributed by the contractor for this CID. These rights extend to Government agencies only, and the data contained in the technical manuals is not to be sold, disclosed or otherwise provided to any other entity or entities outside the Government. The license shall remain in effect as long as the vehicles described in the required technical manuals remain under Government control and usage. In addition, the contractor shall grant permission for the Government to place an Air Force TO (Technical Order) number and publication date as well as distribution, warning, handling and destruction statements on the cover / title pages of the TO.

3.1.1.1.1 Operator's manuals. The operator's manual shall include all information required for the safe and efficient operation of the vehicle, including the fire extinguishing equipment, and any special attachments or auxiliary equipment. The operator's manual shall include at least the following:

a. Location and function of all controls and instruments shall be illustrated and fully described.

b. Safety information that is consistent with the safety standards established by the Occupational Safety and Health Administration (OSHA).

c. Checks and adjustments in preparation for placing the vehicle for service upon receipt from the contractor.

d. Preparation for shipment or storage.

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e. Warranty information and period of the warranty for the complete vehicle and for any component warranty that exceeds the warranty of the complete vehicle. Addresses and telephone numbers shall be provided for all warranty providers.

f. General description of and step-by-step instructions for the operation of the vehicle and its fire extinguishing system(s) and auxiliary equipment.

g. Description of the post-operational procedures (draining, flushing, et cetera).

h. Checklists for the daily maintenance inspection and mission readiness checks that the operator is expected to perform, including basic troubleshooting procedures.

i. Procedures for towing a disabled vehicle.

j. Procedures for changing a tire.

k. Schedules for required preventative maintenance and required periodic maintenance.

l. A line art drawing of the vehicle, including front, rear, left, and right side views, showing basic dimensions and weights (total vehicle and individual axle weight for the unloaded and fully loaded vehicle).

3.1.1.1.2 Service manuals. The service manual shall identify any special tools and test equipment required and shall cover troubleshooting and maintenance as well as minor and major repair procedures. The text shall contain performance specifications, tolerances, and fluid capacities; current, voltage, and resistance data; test procedures; and such illustrations and exploded views as may be required to permit proper maintenance by qualified mechanics. The manual shall contain an alphabetical subject index as well as a table of contents. The service manual shall contain at least the following, where applicable:

a. Fire fighting system schematic(s).

b. Hydraulic schematic.

c. Pneumatic schematic.

d. Electrical schematic.

e. Winterization schematic.

f. Fuel schematic.

g. Schedules for required preventative maintenance and required periodic maintenance.

h. Location, procedure, and interval for parts of the truck and equipment that require lubrication.

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3.1.1.1.3 Parts manual. The parts manual shall include illustrations and exploded views, as needed, to properly identify all parts, assemblies, subassemblies, and special equipment. All components of assemblies shown in illustrations or exploded views shall be identified by reference numbers that correspond to the reference numbers in the parts lists. All purchased parts shall be cross-referenced with the original manufacturer's name and part number. The parts identification manual shall provide the description and quantity of each item used per vehicle. The size, thread dimensions, torque specifications, and special characteristics shall be provided for all nonstandard nuts, bolts, screws, washers, grease fittings, and similar items. The manual shall contain a numerical index. The parts manual shall contain a list of all of the component vendor names, addresses, and telephone numbers referenced in the parts list.

3.1.2 Painting, plating, and corrosion control.

3.1.2.1 Finish. Exterior surfaces shall be prepared, primed, and painted with polyurethane paint in accordance with all of the paint manufacturer's instructions and recommendations. Unless otherwise specified (see 7.2), the exterior finish color shall be Candy Apple Red, Sikkens Color Number FLNA3021, DuPont Color Number 97902U or 4737U, PPG Color Number 71528 (the PPG name for this color is Cardinal Red), or equal. When specified, the exterior finish color shall be Desert Sand, Color Number 30313 of FED-STD-595. The interior of all compartments shall be painted with an impact resistant, textured coating that resists stains, scuffs, chips, and scratches.

a. For vehicles painted Candy Apple Red, the cab upper body (from the bottom of the windshield) and roof shall be painted White, Color Number 17875 of FED-STD-595. All bright metal and anodized parts, such as mirrors, horns, light bezels, treadplate, and roll-up compartment doors, shall not be painted. Compartment interiors shall be Gray.

b. For vehicles painted Desert Sand, all exterior surfaces, excluding all normally bright metal and anodized parts and any interior surfaces visible with any compartment door open (but not the interior of the cab), shall be painted body color. This includes compartment shelves and mounting hardware, but does not include items mounted in the compartments. Non-metallic materials may be black or gray.

3.1.2.2 Dissimilar metals. Dissimilar metals, as defined in MIL-STD-889, shall not be in contact with each other. Metal plating or metal spraying of dissimilar base metals to provide electromotively compatible abutting surfaces is acceptable. The use of dissimilar metals separated by suitable insulating material is permitted, except in systems where bridging of insulation materials by an electrically conductive fluid can occur.

3.1.2.3 Protective coatings. Materials that deteriorate when exposed to sunlight, weather, or operational conditions normally encountered during service shall not be used or shall have a means of protection against such deterioration that does not prevent compliance with performance requirements. Protective coatings that chip, crack, or scale with age or extremes of climatic conditions or when exposed to heat shall not be used.

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3.1.2.4 Reflective stripes. Horizontal reflective stripes shall be applied around the vehicle in accordance with NFPA 414. Bright metal trim or anodized parts may interrupt the reflective stripes. The pattern shall be 10 inches wide with three reflective stripes (1 inch reflective, 1 inch body color, 6 inches reflective, 1 inch body color, and 1 inch reflective). The reflective stripes shall be white for vehicles painted Candy Apple Red and Desert Sand.

3.1.2.5 Lettering. Vehicles painted Candy Apple Red shall have the letters "UNITED STATES" and "AIR FORCE" applied in synthetic or encapsulated gold leaf, with outline and black shadow, on the front door on both sides in long radius elliptical arches above and below the lettering center line. The size of the lettering shall be a minimum of 2½-inches to a maximum of 6-inches. Vehicles painted Desert Sand will not have lettering.

3.1.3 Identification plate. A permanently marked identification plate shall be securely mounted at the driver's compartment. The identification plate shall contain the following information:

a. NOMENCLATURE

b. MANUFACTURER'S MAKE AND MODEL

c. MANUFACTURER'S SERIAL NUMBER

d. USAF VEHICLE REGISTRATION NUMBER

e. NATIONAL STOCK NUMBER (NSN)

f. VEHICLE CURB WEIGHT: kg (pounds)

g. PAYLOAD, MAXIMUM: kg (pounds)

h. GROSS VEHICLE WEIGHT (GVW): kg (pounds)

i. FUEL CAPACITY AND TYPE: gals (gallons) / L (liters)

j. DATE OF DELIVERY (month and year)

k. WARRANTY (months and km (miles))

l. CONTRACT NUMBER

m. PAINT COLOR AND NUMBER

n. LENGTH, WIDTH, AND HEIGHT OF VEHICLE IN INCHES AND CENTIMETERS.

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3.1.3.1 Secondary Identification Plate. A second permanently marked information data plate shall be securely mounted on the interior of the driver's compartment door. The plate shall contain the information required by NFPA 414 paragraph 1.3.5.

3.1.3.2 Combined Identification Plate. A single plate that combines or contains the information required for both plates is acceptable.

3.1.4 Environmental conditions.

3.1.4.1 Temperature range. The vehicle shall be capable of satisfactory storage and operation in temperatures ranging from -40° to 110°F. The vehicle shall be equipped with a cab, chassis, and agent winterization system, permitting operation at -40° F. The winterization system shall not detract from the performance of the vehicle or the firefighting system in ambient temperatures up to 110° F. The vehicle chassis winterization system shall maintain the engine coolant, lubricants, fuel, and electrical systems operational at ambient temperatures of -40° F. The vehicle agent winterization system shall provide sufficient insulation and heating capacity, by means of hot circulating liquids and forced air heat exchangers, to permit satisfactory operation of the vehicle and firefighting systems for a 2-hour period at ambient temperatures as low as -40° F with the vehicle fully operational and the engine running. At the end of this 2-hour period, the vehicle shall be capable of successfully discharging its agents. All compartments not winterized shall be marked on the interior of the compartment as not winterized and not to be used for storage.

3.1.5 Foreign object damage. All loose metal parts, such as pins, shall be securely attached to the vehicle with wire ropes or chains. "Dog tag" style beaded chains shall not be provided. Removable panels, if provided, shall be attached with captive fasteners. Tire valve stem caps shall be made of plastic.

3.1.6 Mobility.

3.1.6.1 Operating terrain. The vehicle shall operate on paved roads, graded gravel roads, cross country terrain, and a sand (desert) environment. Cross country terrain shall consist of open fields, broken ground, and uneven terrain.

3.1.6.2 Gradeability. The fully loaded vehicle shall be able to ascend any paved slope up to and including 50-percent.

3.1.6.3 Side slope stability. The fully loaded vehicle shall be stable on a 30° side slope when tested in accordance with NFPA 414. A slip/trip rail, with a maximum height of 2 inches, may be used. If an adjustable height suspension system is provided, the suspension system may be set to the height normally used on hard pavement.

3.1.6.4 Cornering stability. The fully loaded vehicle shall be stable in accordance with NFPA 414 when tested in accordance with NFPA 414.

3.2 Weights and dimensions.

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3.2.1 Overall dimensions. Overall dimensions shall be the minimum consistent with the operational performance and the design constraints necessary to achieve the specified performance. Overall dimensions shall not exceed:

	1500 Gallon	3000 Gallon
Length (inches)	420	480
Width (inches, excluding mirrors)	120	120
Height (inches)	163	163

3.2.2 Angles of approach and departure. The fully loaded vehicle shall have angles of approach and departure of not less than 30°.

3.2.3 Field of vision. The vehicle shall have a field of vision in accordance with NFPA 414.

3.2.3.1 Mirrors. Combination flat and convex outside rearview mirrors shall be installed on each side of the cab. The flat mirrors shall be of the motorized remote control type, providing not less than 60° horizontal rotational viewing range. The flat mirrors shall also have electrically heated heads. Mirror remote and heating controls shall be located on the instrument panel within reach of the seated driver. To provide the driver a clear view of the area ahead of the vehicle, a rectangular mirror shall be installed on the lower corner of each side of the windshield, but not within the driver's direct line of sight, having a minimum area of 35 square inches.

3.3 Chassis and vehicle components.

3.3.1 Engine. The vehicle shall have a turbocharged diesel engine that is certified to comply with the Environmental Protection Agency (EPA) on- or off-highway emission requirements at the time of manufacture.

3.3.1.1 High idle switch. A high idle switch, which does not increase engine speed more than 25 percent above normal low idle speed and does not exceed the engine manufacturer's recommendation, may be provided to increase alternator, air compressor, or air conditioning compressor output to meet the maximum load requirements. A lighted rocker or toggle switch, accessible from the driver's seated position, shall activate the high idle control unit. The high idle switch shall operate only when the vehicle is out of gear and the parking brake is engaged. The high idle control unit shall automatically disengage if the transmission is placed in gear, the parking brake is released, or the fire pump is engaged. The high idle control unit shall not engage automatically; the high idle control unit shall only engage when the high idle switch is in the "ON" position.

3.3.1.2 Acceleration. The fully loaded vehicle shall accelerate from 0 to 50 miles per hour (mph) within 25 seconds (1500 gallon truck) or 35 seconds (3000 gallon truck) on a level, paved road.

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3.3.1.3 Maximum speed. The fully loaded vehicle shall attain a minimum top speed of 70 mph on a level, paved road.

3.3.1.4 Pump and roll on a 40-percent grade. The fully loaded vehicle shall be capable of pump and roll operations on a paved, dry, 40-percent grade in accordance with NFPA 414.

3.3.2 Engine cooling system. The engine cooling system shall be in accordance with NFPA 414. Silicone radiator and heater hoses, constant-torque clamps, and a coolant filter shall be provided. A label shall be installed near the engine coolant reservoir reading “Engine Coolant Fill.”

3.3.3 Fuel system. The fuel system shall be in accordance with NFPA 414.

3.3.3.1 Fuel priming pump. The vehicle shall be equipped with an electric fuel pump in addition to the mechanical fuel pump. The electric pump shall be used as a priming pump capable of re-priming the engines fuel system.

3.3.3.2 Fuel filters. Primary and secondary fuel filters and a heated fuel/water separator shall be provided. The fuel/water separator shall include a water coalescer and a drain valve that is readily accessible by an operator or a mechanic, and shall be in accordance with SAE J1839. A combination fuel filter and fuel/water separator may be provided. Fuel filter elements shall be easily replaceable by a mechanic without loss of engine prime.

3.3.3.3 Fuel tank. The vehicle shall have one or two fuel tanks with a minimum usable capacity in accordance with NFPA 414. Each tank shall have a fill opening of 3 inches minimum, readily accessible to personnel standing on the ground and designed to prevent fuel splash while refueling. Each tank shall be located and mounted so as to provide maximum protection from damage, exhaust heat, and ground fires. If more than one tank is furnished, means shall be provided to assure equalized fuel level in both tanks. An overturn fuel valve shall be provided for each tank to prevent spillage in the event of a rollover. Each fuel tank must be prominently labeled “Diesel Fuel Only”.

3.3.4 Exhaust system. The exhaust system shall be in accordance with NFPA 414. The exhaust system including the muffler shall be constructed of stainless steel. Exhaust system outlet(s) shall be directed upward or to the rear, away from personnel accessing equipment compartments and the engine air intake, and shall not be directed toward the ground.

3.3.5 Transmission. A fully automatic transmission with a hydraulic torque converter and at least five forward speeds shall be provided. The normal driving range selector position shall provide at least four gear ratios without movement of the selector. The transmission shall be in accordance with NFPA 414.

3.3.6 Driveline. The vehicle driveline shall be in accordance with NFPA 414. If the driveline is equipped with a differential locking control, a warning/caution label shall be placed in view of the driver indicating the proper differential locking/un-locking procedures. The operator’s manual shall also include a similar warning/caution.

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3.3.7 Axle capacity. Each axle shall have a rated capacity, as established by the axle manufacturer, in accordance with NFPA 414.

3.3.8 Suspension. The suspension system shall be in accordance with NFPA 414.

3.3.9 Tires and wheels. Tires and wheels shall be in accordance with NFPA 414. The vehicle shall be equipped with single tires and wheels at all wheel positions. The vehicle shall be equipped with tubeless steel belted radial tires with non-directional on/off-road type tread mounted on disc wheel assemblies. Tire and wheel assemblies shall be identical at all positions. Tires and wheels shall be certified by the manufacturer for not less than 25 miles of continuous operation at 60 mph at the normal operational inflation pressure. A spare tire and wheel assembly shall be provided; however, the spare tire and wheel assembly is not required to be mounted on the vehicle. Tires shall be new. Retreads recaps, or re-grooved tires shall not be permitted. Tire valve stem caps shall be made of plastic.

3.3.10 Towing connections. The vehicle shall be equipped with towing connections in accordance with NFPA 414. The vehicle shall be designed for flat towing; the capability to lift and tow the vehicle is not required.

3.3.11 Brake system. The vehicle shall be equipped with a multi-channel all-wheel antilock brake system with at least one channel per axle;. The brakes shall be fully air-actuated; outboard mounted disk type. Brakes shall be in accordance with Code of Federal Regulations (CFR) 49 CFR 393.40 through 393.42(b)), 393.43, and 393.43 through 393.52. The braking system complete with all necessary components shall include:

a. Air compressor having a capacity of not less than 16 standard cubic feet per minute (scfm).

b. Air storage reservoir(s), each tank equipped with drain, and with safety and check valves between the compressor and the reservoir tank.

c. Automatic moisture ejector on each air storage reservoir. Manual air tank drains are an acceptable alternative if they are labeled and are located in one compartment, accessible by an individual standing at the side of the vehicle.

d. Automatic slack adjusters on cam brakes or internal self-adjusting brakes on wedge brakes on all axles.

e. Spring set parking brakes.

f. All components of the braking system shall be installed in such a manner as to provide adequate road clearance when traveling over uneven or rough terrain, including objects liable to strike and cause damage to the brake system components. No part of the braking system shall extend below the bottom of wheel rims, to ensure, in case of a flat tire, that the weight of the vehicle will be supported by the rim and the flat tire and not be imposed

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on any component of the braking system. Slack adjusters and air chambers shall be located above the bottom edge of the axle carrier.

3.3.11.1 Air dryer. A replaceable cartridge desiccant air dryer shall be installed in the air brake system. The dryer shall have the capability of removing not less than 95 percent of the moisture in the air being dried. The dryer shall have a filter to screen out oil and solid contaminants. The dryer shall have an automatic self-cleaning cycle and a thermostatically controlled heater to prevent icing of the purge valve.

3.3.11.2 Compressed air shoreline. A checked, auto-eject compressed air shoreline connection shall be provided to maintain brake system pressure while the vehicle is not running. The shoreline shall be flush mounted (not to extend outside the body line), located on the exterior of the vehicle at the rear. A minimum 50 foot long air supply hose equipped with an appropriate mating shoreline connector and an air fitting shall be provided with the vehicle.

3.3.12 Steering. The vehicle shall be equipped with power steering. A mechanical steering system shall be installed on the rear axle of the tandem axle on a 6x6 (3000 gallon) vehicle to reduce the vehicle's turning circle while minimizing tire scrub on that axle.

3.3.12.1 Steering effort. The steering system performance shall be in accordance with NFPA 414.

3.3.12.2 Turning diameter. The fully loaded vehicle shall have a wall to wall turning diameter of less than three times the overall length of the vehicle in both directions.

3.3.13 License plate bracket. A lighted license plate bracket shall be provided at the left rear of the vehicle.

3.4 Cab. The vehicle shall have a fully enclosed two door cab of corrosion resistant construction. Cab door openings shall extend for the full vertical height of the side panels. Steps and handrails shall be provided for all crew doors, and at least one grab handle shall be provided for each crew member, located inside the cab for use while the vehicle is in motion. The lowermost step(s) shall be no more than 22 inches above level ground when the vehicle is fully loaded. A tilt and telescoping steering column shall be provided.

3.4.1 Windshield and windows. The windshield and windows shall be of tinted safety glass. Each door window shall be capable of being opened far enough to facilitate emergency occupant escape in the event of a vehicle accident. If the cab is equipped with power windows, a roof hatch shall be provided to facilitate emergency occupant escape in the event of a vehicle accident. If a roof escape hatch is provided, a label reading "Emergency Escape Hatch" shall be installed on the cab interior side of the hatch.

3.4.2 Cab interior sound level. The maximum cab interior sound level shall be in accordance with NFPA 414.

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3.4.3 Instruments and controls. All instruments and controls shall be illuminated. Gauges shall be provided for engine oil pressure, engine coolant temperature, and automatic transmission oil temperature. All switches and/or controls that activate systems or system components shall be labeled as to their function. In addition to the instruments and controls required by NFPA 414, the following shall be provided within convenient reach of the seated driver:

a. Master warning light control switch,

b. Work light switch(es),

c. Compartment "Door Open" warning light and intermittent alarm that sounds when a compartment door is open and the parking brakes are released or the transmission is in any position other than neutral and

d. An engine hour meter.

3.4.4 Windshield deluge system. The vehicle shall be equipped with a powered windshield deluge system. The deluge system shall be supplied from the agent water tank and shall have an independent pumping system. The deluge system activation switch shall be located within reach of the seated driver and turret operator. The deluge system shall include a replaceable filter and a manual valve for use when the filter is changed.

3.4.5 Forward looking infrared. The vehicle shall be equipped with a forward looking infrared (FLIR) system in accordance with NFPA 414. The FLIR camera shall incorporate microbolometer detection technology. The FLIR monitor described in NFPA 414 shall have a minimum dimension of 10 inches (measured diagonally). The monitor shall be so located as to be visible to both the seated driver and turret operator.

3.4.6 Climate control system. The contractor's standard heater/defroster and air conditioning system shall be provided. In 100° F ambient temperature and at maximum compressor speed, the air conditioning system shall cool the fully occupied cab to 75° F within 30 minutes. The climate control system shall induct at least 60 cubic feet per minute of fresh air into the cab. At least four adjustable panel louvers shall be provided, approximately evenly spaced across the width of the cab; each shall provide approximately equal flow. Cab mounted components shall be protected from inadvertent damage by personnel.

3.4.7 Seats. The driver seat and the turret operator’s seat to the driver’s right shall each be air suspension type, capable of being adjusted fore and aft and for height. The driver’s seat shall also be equipped with a recline feature. A third fixed (non-suspension) seat for an additional crew member shall be located to the left, and possibly behind the driver. The turret operator and the crew member’s seats shall both be provided with a backrest and a bracket designed to store a 1-hour capacity self-contained breathing apparatus (SCBA). Each seat shall be provided with a Type 2 seat belt assembly (i.e., 3-point restraint) in accordance with Code of Federal Regulations (CFR) 49 CFR 571.209. A seat belt which is an integral part of the seat with no remote mounting point shall be provided. Seat belts must be of sufficient length to accommodate personnel in full personal protective equipment (PPE).

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3.4.8 Windshield wipers and washer. The vehicle shall be equipped with electrically powered windshield wipers. The wiper arms and blades shall be of sufficient length to clear the windshield area described by SAE J198. Individual wiper controls shall include a minimum of two speed settings and an intermittent setting. The wiper blades shall automatically return to a park position, out of the line of vision. The vehicle shall be equipped with a powered windshield washer system, including an electric fluid pump, a minimum one gallon fluid container, washer nozzles mounted to the wiper arms (wet arms), and a momentary switch.

3.4.9 Warning signs. Signs that state "Occupants must be seated and wearing a seat belt when apparatus is in motion" shall be provided in locations that are visible from each seated position in accordance with NFPA 414."

3.4.10 Lateral accelerometer. The vehicle shall be equipped with a lateral accelerometer in accordance with NFPA 414.

3.5 Body, compartments, and equipment mounting.

3.5.1 Body. The vehicle shall have a corrosion resistant body.

3.5.2 Compartments. The vehicle body shall have lighted compartments in accordance with NFPA 414 with a minimum of 10 cubic feet of enclosed storage space.

3.5.2.1 Compartment doors. Storage compartments shall have clear anodized aluminum, counterbalanced, non-locking, roll-up doors. Door latch handles shall be full-width bar type. Door straps shall be provided to assist in closing the compartment doors when the rolled up or hinged door height exceeds six feet above the ground.

3.5.2.2 Scuffplates. Replaceable scuffplates shall be provided at each compartment threshold to prevent body damage from sliding equipment in and out of the compartments. The scuffplates shall be securely attached to the compartment threshold but shall be easily replaceable in the event of damage.

3.5.2.3 Drip rails. Drip rails shall be provided over each compartment door. If the drip rails are not integral with the body, they shall be of anodized extruded aluminum and shall have a bright finish for vehicles painted Candy Apple Red.

3.5.2.4 Shelves. An adjustable and removable compartment shelf shall be provided for every 18 inches of each vertical storage compartment door opening. Shelving adjustments shall require no more than common hand tools, and shall not require disassembly of fasteners. Shelves shall support a minimum of 500 pounds without permanent deformation. Each shelf shall be accessible to crew members standing on the ground. A pull out and tip-down configuration shall be used for shelving over 54 inches from the ground when the compartment is more than 18 inches deep. Each shelf shall have ¼ inch diameter drain holes located in each corner of the shelves so as to allow for drainage of any water from the stowed equipment.

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3.5.2.5 Drainage mats. Each compartment floor and shelf shall be covered with a removable black mat designed to allow for drainage of any water from the stowed equipment.

3.5.3 SCBA storage tubes. Tubes for storage of four SCBA 1-hour bottles shall be provided, two on each side of the vehicle. The tubes shall be in accordance with NFPA 1901 and shall be of sufficient size to accommodate the SCBA cylinders.

3.5.4 Ladder, handrails, and walkways. Ladders, stepping, standing, and walking surfaces shall be in accordance with NFPA 414. Handrails shall be provided in accordance with NFPA 414. The lowermost step(s) or ladder rungs shall be no more than 22 inches above level ground when the vehicle is fully loaded.

3.5.5 Ancillary equipment. The following equipment items shall be furnished. These items will be inventoried, packaged, and shipped as loose equipment. Mounting locations shall be determined and accomplished by the receiving installation.

a. Two spanner wrench holders.

b. Four spanner wrenches.

c. Two hydrant wrench holders.

d. Two hydrant wrenches.

e. One rubber mallet, 24 ounce, 15-inch overall length, 2-inch diameter, 4-inch long head.

f. Two couplings, hose, double male, swivel, internal 2½-inch hard coated aluminum with rocker lug.

g. Two adapters, double female, swivel, 2½-inch, aluminum.

h. One 2½-inch to two 1½-inch gated wye connection.

i. One 20 foot length of 6-inch soft suction hose with 4½-inch female National Hose thread connections.

j. One axe with a fiberglass handle, flathead and blade shield and mounting bracket.

k. One axe with a fiberglass handle, pickhead and blade shield and mounting bracket.

l. One extinguisher, 2½-gallon pressurized water and mounting bracket.

m. One extinguisher, 20 pound, dry chemical and mounting bracket.

n. One set of folding wheel chocks and mounting brackets.

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o. Storz wrenches and holders, 4-inch to 4½-inch.

p. Emergency road kit consisting of at least the following items:
1. Two highway warning triangles.
2. Two 20 minute flares.
3. One reflective vest.
4. Two cyalume light sticks.
5. One water resistant storage bag.

3.6 Agent system.

3.6.1 Agent pump. The vehicle shall be equipped with a pump capable of providing the performance specified herein.

3.6.1.1 Agent system piping. All metallic surfaces of the piping and associated components that come into contact with the agent shall be of brass, bronze, or passivated stainless steel.

3.6.1.2 Tank to pump connection. A check valve and shutoff valve shall be provided in each tank to pump line.

3.6.1.3 Intake connections. The vehicle shall be equipped with one valved 4½-inch intake connection on the left side. The vehicle shall be equipped with one valved 2½-inch intake connection on the left side, adjacent to the 4½-inch intake connection and fitted with either 30° or 45° turn-down fittings. The 4½-inch intake connection shall have male National Hose threads, a quarter-turn control valve, a bleeder valve, a strainer, a cap, and a slow-operating valve and an automatic pressure relief device in accordance with NFPA 1901. The 2½-inch intake connection shall have rocker lug female National Hose threads, a quarter-turn control valve, a bleeder valve, a strainer, and a plug in accordance with NFPA 1901. The vehicle shall be capable of filling its water tank by pumping from a draft, a hydrant, or a nurse truck through any of the intake connections without the use of a hose from a discharge connection to a tank fill connection.

3.6.1.4 Discharge connections. Two 2½-inch discharge connections with male National Hose threads shall be provided. One 2 ½ - inch discharge shall be provided on each side of the vehicle. Each connection shall be equipped with a cap, a quarter-turn control valve, a bleeder valve, and a pressure gauge. Each connection shall be rated at 250-gpm minimum.

3.6.1.5 Piping, couplings, and valves. The agent system piping shall incorporate groove type couplings to the maximum practical extent. All piping sections between agent system components shall be readily removable without disturbing the components.

3.6.1.6 Overheat protection. The agent system shall be equipped with an overheat protection system in accordance with NFPA 414.

3.6.1.7 Pressure relief valves. The agent system shall be equipped with pressure relief valves in accordance with NFPA 414.

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3.6.1.8 Drains. The agent system shall be equipped with a drainage system in accordance with NFPA 414.

3.6.1.9 Priming pump. The vehicle shall be equipped with a priming pump.

3.6.2 Water tank. The vehicle shall have a water tank with a manufacturer certified minimum capacity of at least 1,500 or 3000 gallons.

3.6.2.1 Water tank construction. The water tank shall be constructed of materials to meet the requirements of NFPA 414 and shall be supplied with a lifetime warranty. All materials used shall be capable of storing water, foam concentrate, and water/foam solutions.

3.6.2.2 Water tank manhole cover and drain. The water tank shall be equipped with an easily removable manhole cover with a minimum opening diameter of 20 inches. The water tank shall incorporate a drain and drain valve. The valve shall be on the left side of the vehicle and controlled by a crew member standing on the ground. The drain line shall be 2-inch internal diameter (I.D.) minimum. The point for discharge for the water tank drain shall be below the under body panels.

3.6.2.3 Water tank overflow system and venting. The water tank shall incorporate a vent system to relieve pressure on the tank during fill and discharge operations at maximum flow rates and an overflow system to relieve excess liquid in the event of tank overfill. The vent and overflow system shall prevent leakage of water when the tank is filled to capacity and the vehicle is operating on the maximum side slopes and grades specified herein. Drainage from the vent and overflow system shall not flow over body panels or other vehicle components and shall not be in the track of any of the tires. Tank vent hoses shall be of the non-collapsible type.

3.6.2.4 Water tank top fill opening. A top fill opening of not less than 8 inches internal diameter with a readily removable ¼-inch mesh strainer shall be provided. The fill opening may be incorporated as part of the manhole cover, and shall be sized to accommodate a 2½-inch fill hose. The opening shall be marked “Water Fill.”

3.6.2.5 Water tank fill connections. The water tank shall incorporate a 4½-inch male National Hose thread connection and a 2½-inch rocker lug female National Hose thread connection on each side of the vehicle. Each connection shall be fitted with a 30° turn-down fitting. The water fill shall allow external re-supply of the water tank during discharge pumping operations. Each tank fill connection shall be in accordance with NFPA 414.

3.6.3 Foam system.

3.6.3.1 Foam concentrate tank. The foam concentrate tank(s) shall have a manufacturer certified working capacity sufficient for two tanks of water at the maximum tolerance specified in NFPA 412, *Standard for Evaluating Aircraft Rescue and Fire-Fighting Foam Equipment* for 6-percent foam concentrate (i.e., 7.0-percent).

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3.6.3.1.1 Foam tank construction. The foam tank shall be constructed of materials to meet the requirements of NFPA 414 and shall be supplied with a lifetime warranty. All materials used shall be capable of storing foam concentrate.

3.6.3.1.2 Foam tank drain. The foam tank shall incorporate a drain and drain valve. The valve shall be on the left side of the vehicle and controlled by a crew member standing on the ground. The drain line shall have a minimum 1½-inch I.D. The foam tank drain outlet shall be located so that the contents of the tank can be drained into 5-gallon cans and 55-gallon drums.

3.6.3.1.3 Foam tank top fill trough. The foam tank shall incorporate a top fill trough mounted in the top of the tank readily accessible to at least two crew members on top of the vehicle. The top fill trough shall incorporate a cover and latch, sealed so as to prevent spillage under any operating condition. The top fill trough shall be designed to allow two standard 5-gallon foam concentrate containers to be emptied simultaneously. The top fill trough neck shall extend sufficiently close to the bottom of the tank to reduce foaming to a minimum during the fill operation. The top fill trough shall incorporate readily removable, rigidly constructed 10 mesh stainless steel or brass strainers. All components in and around the top fill trough shall be constructed of materials that resist all forms of deterioration that could be caused by the foam concentrate or water. The opening shall be marked “Foam Fill.”

3.6.3.2 Foam tank fill connections. The foam tank shall incorporate a 1½-inch National Hose thread female hose connection on each side of the vehicle to permit filling by an external transfer hose at flow rates up to 75-gpm. The connections shall be provided with chained-on long handled plugs or rocker lug plugs. The top of the connections shall be no higher than 48 inches above the ground and readily accessible. The fill lines shall incorporate check valves and readily removable, rigidly constructed ¼-inch mesh strainers. All components in the foam tank fill system shall be constructed of materials that resist all forms of deterioration that could be caused by the foam concentrate or water.

3.6.3.2.1 Foam tank vent and overflow system. The foam tank shall incorporate a vent system to relieve pressure on the tank during fill and discharge operations at maximum flow rates and an overflow system to relieve excess liquid in the event of tank overfill. The vent and overflow system shall prevent leakage of foam when the tank is filled to capacity and the vehicle is operating on the maximum side slopes and grades specified herein. Drainage from the vent and overflow system shall not flow over body panels or other vehicle components and shall not be in front of or behind any of the tires. Tank vent hoses shall be of the non-collapsible type.

3.6.3.3 Foam transfer pump. An electric motor driven or pneumatic, self-priming, diaphragm pump shall be provided and mounted in a compartment on the vehicle. The pump shall be driven from the vehicle electrical or air system. The pump shall be capable of transferring and drawing foam liquid concentrate at adjustable flow rates up to 25-gpm directly through the pump and loading connections (see 3.6.3.2). All materials and components that come in contact with the foam shall be compatible with the foam concentrate. The pump and its plumbing shall have provisions for flushing with water from the water tank. The pump shall be removable as an assembly without disturbing other components. A suitable length of hose with appropriate connections shall be provided for filling the foam tank from an external foam storage container.

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3.6.3.4 Foam flushing system. The foam concentrate system shall be designed in accordance with NFPA 414 so that the system can be readily flushed with clear water.

3.6.3.5 Foam concentrate piping. All metallic surfaces of the piping and associated components that come into contact with the foam concentrate shall be of brass, bronze, or passivated stainless steel. The foam concentrate piping shall be in accordance with NFPA 414.

3.6.4 Foam proportioning system. The vehicle shall have an electronic foam proportioning system for use with 3% Aqueous Film-Forming Foam (AFFF). The system shall automatically and uniformly proportion water and 3% foam concentrate at any ratio from 0-10%.

3.6.5 Primary turret. Unless otherwise specified (see 7.2), the vehicle shall be equipped with a roof turret in accordance with 3.6.5.1. When specified, the vehicle shall be equipped with an extendable turret in accordance with 3.6.5.2.

3.6.5.1 Roof turret. The roof turret shall be mounted near the front of the roof at the center of the vehicle. It shall have a non-air-aspirating, constant flow, variable stream nozzle with dual flow rates for foam or water rated at 750-gpm (1500) 1,250-gpm (3000) minimum. The discharge pattern shall be infinitely variable from straight stream to fully dispersed. The roof turret shall be power operated; power controls shall be positioned for use by the driver and the crew member seated to the right of the driver. The nozzle shall be designed to drain automatically when the discharge valve is in the OFF position.

3.6.5.2 Extendable Turret. The extendable turret shall be in accordance with NFPA 414 and shall have the vertical and horizontal reach necessary to service the highest placed engine of the aircraft being serviced (DC-10/KC-10). It shall have a non-air-aspirating, constant flow, variable stream nozzle with dual flow rates for foam or water rated at 1,000-gpm minimum. The discharge pattern shall be infinitely variable from straight stream to fully dispersed. The nozzle shall be designed to drain automatically when the discharge valve is in the OFF position. The extendable turret shall be controlled by one or two joysticks, each with a pistol grip handle, positioned for use by the driver and the crew member seated to the right of the driver (the turret operator). The cab design shall provide clear visibility of the turret to both the driver and the turret operator with the turret in any position.

3.6.5.2.1 Video camera and monitor. When an extendable turret is specified (see 7.2), it shall be equipped with a remote video camera and a cab mounted monitor. The system shall be a complete video system consisting of a single color camera equipped with auto-focus and a cab controlled zoom. The camera/lens assembly shall be protected from the heat of the fire and from the same climatic extremes as the truck. A color video monitor with a minimum dimension of 10 inches (measured diagonally) shall be positioned in the cab within view of both the driver's and the turret operator's seated positions. One monitor may be provided for both the FLIR camera (see 3.4.5) and the video camera with a switch to change between cameras.

3.6.5.2.2 Aircraft skin penetrator. When an extendable turret is specified (see 7.2), it shall be equipped with an aircraft skin penetrator and agent application tool. The skin penetrator shall be

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a minimum of 24 inches long, installed at the tip of the extendable turret, and connected to the water/foam agent discharge line. Agent application through the skin penetrator shall be controlled from the cab.

3.6.6 Bumper turret. The vehicle shall be equipped with a joystick controlled, constant flow, non-air-aspirating, variable stream type bumper turret. The bumper turret shall be capable of discharging at a flow rate of 300-gpm minimum of foam or water, with a pattern infinitely variable from straight stream to fully dispersed. The nozzle pressure for the bumper turret shall be 1300 psi. The bumper turret shall be capable of automatic oscillation, with the range of oscillation fixed at 45° each side of center (left and right) and elevation/depression.

3.6.7 Preconnected handlines. A 200 foot, 1¾-inch preconnected woven jacket handline, with a 1½-inch control valve and a pistol grip nozzle, shall be located on each side of the vehicle. A safety system shall be provided to prevent charging of the hose until the hose has been fully deployed. The handlines and nozzles shall be in accordance with NFPA 414. A control for charging each handline shall be provided for operation by both the driver and the turret operator.

3.6.8 Structural panel. The vehicle shall be equipped with an agent system structural control panel, on the left side of the vehicle, operable while standing on the ground. Structural panel activation shall be interlocked to operate only with the vehicle parking brakes set and the transmission in neutral position. Controls and instruments shall be grouped by function. The control panel shall be hinged or accessible from the rear for maintenance. Instruments shall be lighted for night operation. The structural panel shall include, as a minimum, the following:

a. Panel activation switch, including the panel lights.

b. Manual engine throttle with micro adjustments, or may be pilot controlled by manually regulated agent system pressure.

c. Engine tachometer.

d. Engine oil pressure gauge with low pressure warning light.

e. Engine coolant temperature gauge with high temperature warning light.

f. Liquid filled pump suction gauge, -30 inches Hg vacuum to 600 psi.

g. Liquid filled pump pressure gauge, 0 to 600 psi.

h. Manually adjustable pump pressure, using either a relief valve with indicator lights, pilot controlled engine throttle, or an electronic pressure sensing governor.

i. Foam or water selection switch.

j. Primer pump control.

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k. Water and foam tank liquid level indicators. Identical water and foam tank liquid level indicators shall be provided on the both sides of the vehicle, adjacent to the water and foam tank fills.

l. Two 10 foot long, light weight 5-inch hard suction hoses, with 4½-inch National Hose thread long handle female couplers on one end and 4½-inch National Hose thread long handle male couplers on the other, shall be carried on the top of the vehicle or stored in a recessed compartment in drainable trays. A chrome-plated 4 ½ -inch National Hose thread barrel suction strainer and mounting hardware shall be provided with the vehicle and stored in a compartment.

3.7 Dry chemical agent system. The vehicle shall be equipped with a 500-pound minimum capacity potassium bicarbonate dry chemical auxiliary agent system. The propellant gas cylinder shall be replaceable within fifteen minutes by two crew members standing on the ground. The propellant gas cylinder shall be secured to withstand off-road operations. A pressure indicator shall be visible to any person opening the tank fill cap. Blow-down piping shall be directed beneath the vehicle. The dry chemical agent tank shall include lifting rings and shall have a nameplate indicating, as a minimum, the following:

a. Extinguishing agent.

b. Capacity.

c. Weight full.

d. Weight empty.

e. Operating pressure.

f. Hydrostatic test date.

g. Type of agent required for re-servicing.

3.7.1 Dry chemical primary turret discharge nozzle. The vehicle shall be equipped with a combination dry chemical/AFFF turret of the concentric direct injection type, designed to entrain the dry chemical agent within the foam solution discharge. Dry chemical discharge control shall be within reach of the driver and the turret operator.

3.7.2 Dry chemical hose reel. A hose reel, equipped with at least 150 feet of dry chemical hose, shall be mounted in a compartment. The handline agent and purge controls shall be mounted in or adjacent to the compartment. All electrical components shall be sealed against entry of water. The hose reel shall have both electric and manual rewind provisions. The manual rewind handle shall be bracket mounted and stored in the compartment. A quick acting control shall be provided to activate the handline from the cab of the vehicle.

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3.8 Electrical systems and warning devices. The vehicle shall have an electrical and starting system in accordance with NFPA 414.

3.8.1 Alternator. A single or dual alternator charging system in accordance with NFPA 414 shall be provided. The minimum continuous electrical load shall include operation of the air conditioning system.

3.8.2 Batteries. Batteries shall be of the maintenance-free type; addition of water shall not be required during normal service life. The battery cover and vent system shall be designed to prevent electrolyte loss during service and to keep the top of the battery free from electrolyte.

3.8.2.1 Battery compartment. The batteries shall be enclosed in a weatherproof box or compartment and be readily accessible.

3.8.3 Battery charger or conditioner. The vehicle shall have a DC taper type battery charger or an automatic battery conditioner, providing a minimum 12 amp output. The charger/conditioner shall be permanently mounted on the vehicle in a properly ventilated, accessible location. The charger/conditioner shall be powered from the electrical shoreline receptacle (see 3.9.1). A charging indicator shall be installed next to the receptacle. When a battery conditioner is provided, the conditioner shall monitor the battery state of charge and, as necessary, automatically charge or maintain the batteries without gassing, depleting fluid level, overheating, or overcharging. A slave receptacle shall be provided at the rear of the vehicle.

3.8.4 Electromagnetic interference. The vehicle electrical system shall be in accordance with IEC CISPR 12 and IEC CISPR 25 for electromagnetic interference.

3.8.5 Work lighting.

3.8.5.1 Cab interior lights. Cab interior light levels shall be sufficient for reading maps or manuals. At least one red and one white cab interior dome light shall be provided.

3.8.5.2 Compartment lights. LED strip lighting shall be placed along each side of a compartment equipped with a roller shutter door. Compartments having top or side hinged doors shall be provided with circular LED lights. Where a shelf is provided, this illumination shall be provided both above and below the shelf. Lights shall automatically illuminate only when the respective door is opened and the vehicle electrical master switch is in the "on" position. Light switches shall be of the magnetic (non-mechanical) type.

3.8.5.3 Ladder, step, walkway, and area lights. Non-glare white or amber LED lighting shall be provided at ladders and access steps where personnel work or climb during night operations. In addition, ground lighting shall be provided. These area lights shall be controlled with three-way switches on the cab instrument panel and near the light sources. The switch located in the cab shall be a master switch and must be turned on before auxiliary switches near the light sources are operational.

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3.8.5.4 Floodlights. Two halogen spot/floodlight(s) shall be attached at the end of the roof turret or at the end of the extendable turret assembly. The spot/floodlight(s) shall illuminate the area covered by the turret or HRET. The spot/floodlight(s) shall be switched from the instrument panel.

3.8.5.5 Scene lights. A total of six LED high mounted floodlights shall be provided to illuminate the work areas around the vehicle: two on each side and two in the front. Each pair of side mounted lights shall be controlled by a switch mounted on the side of the vehicle and by a switch mounted on the instrument panel. A switch shall also be mounted on the instrument panel to control the lights at the front.

3.8.6 Audible warning devices.

3.8.6.1 Siren. The vehicle shall be equipped with an electronic siren system. The amplifier unit shall include volume control and selection of "Radio," "PA," "Manual," "Yelp," "Wail," and "Hi-Lo" (European) modes, and a magnetic noise canceling microphone. The amplifier, microphone, and controls shall be within reach of the driver and the turret operator. Siren activating foot switches shall be located in front of the driver and the turret operator. The siren speaker shall be rated at 100 watts minimum and shall be located in a guarded position as low and as far forward on the vehicle as practical.

3.8.6.2 Horn. Dual forward facing air horns shall be installed in protected locations near the front of the vehicle. Air horn activating foot switches shall be located in front of the driver and the turret operator.

3.8.7 Emergency warning lights. All emergency warning lights shall use light emitting diode (LED) elements. Three identical LED light bars shall be provided, with two mounted on top of the vehicle, one toward the outside of the vehicle on each side, on or immediately behind the cab. Each of these light bars shall have twelve LED elements, one on each corner, three at the front, three at the rear and one on each side. A third LED light bar shall be mounted on the top rear center of the vehicle. This light bar shall also have eight twelve LED elements, one on each corner, three at the front, three at the rear and one on each side. Three LED warning lights shall be provided on each side, one mounted as far forward, one mounted as far to the rear and one mounted as close as practical to the mid-point of the vehicle. Two LED warning lights shall be mounted on the front of the vehicle. Two LED warning lights shall be mounted on the rear of the vehicle. A switch shall be provided on the instrument panel to control all of the top, side, front and rear emergency warning lights. A switch shall also be provided on the instrument panel to disable all ten lower emergency warning lights when desired.

3.8.7.1 Emergency warning light color. Unless otherwise specified, all emergency warning lights shall be red. When specified (see 7.2), the rearward, red emergency warning lights shall be replaced with amber. When specified (see 7.2), all red emergency warning lights shall be replaced with blue.

3.8.7.2 Headlight flashing system. A high beam, alternating/flashing, headlight system shall be provided. The headlight flasher shall be separately switched from the warning light panel.

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3.8.8 Radio circuit. The vehicle shall have two separate 30 amp circuits, with breakers and at least 6 feet of wire routed to a space provided adjacent to the driver and turret operator for purchaser provided radios and other electrical equipment. The wiring shall be tagged indicating its purpose. Two antennas shall be installed on the roof of the cab, pre-wired to the space in the cab in the cab where purchaser provided radios would be installed.

3.8.9 Auxiliary power receptacles. The vehicle shall have 2 12-volt auxiliary power receptacles mounted adjacent to the driver and crew member positions, preferably in the instrument panel.

3.9 Line voltage electrical system.

3.9.1 Electrical shoreline connection. The battery charger/conditioner shall be powered from a covered, polarized, insulated, labeled, recessed, male, 110 volt AC auto-eject receptacle. The connection shall be located on the exterior of the vehicle at the rear. A weatherproof charge meter shall be installed next to the receptacle. A 50 foot long, three wire, 15 amp rated, 110 volt, AC power cable, with straight blade (non twist-lock) connectors, shall be provided. When specified (see 7.2), the receptacle and power cable shall be for 220 volts in lieu of 110 volts.

3.10 Air systems.

3.10.1 Air hose reel. An air hose reel shall be provided in a compartment on the side of the vehicle. The hose reel shall be equipped with 200 feet of 3/8-inch I.D. hoseline. A 3/8-inch National Pipe Taper (NPT) fitting and female style quick disconnect shall be connected to the end of the hoseline. A four-way roller guide shall be provided for the hose reel to prevent hose chafing and kinking. The hoseline shall be equipped with a rubber ball stop to prevent hose pull through on roller guides during rewinding operations. The hose reel shall have an electric rewind motor and provisions for manual rewind in the event of motor failure; the manual rewind handle shall be securely stored near the hose reel. A pressure protected air supply from the chassis air system shall be connected to the hose reel. The air supply lines shall be routed with minimum bends and located or guarded from damage from the carried equipment.

3.11 Workmanship. The vehicle, including all parts and accessories, shall be fabricated in a thoroughly workmanlike manner. Particular attention shall be given to freedom from blemishes, burrs, defects, and sharp edges; accuracy of dimensions, radii of fillets, and marking of parts and assemblies; thoroughness of welding, brazing, soldering, riveting, and painting; alignment of parts; tightness of fasteners; et cetera. The vehicle shall be thoroughly cleaned of all foreign matter.

4.1 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs. The contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with 23.403 of the Federal Acquisition Regulation (FAR). However, used, rebuilt, or refurbished items shall not be provided.

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4.2 Green Procurement Program. Green Procurement Program (GPP) is a mandatory federal acquisition program that focuses on the purchase and use of environmentally preferable products and services. GPP requirements apply to all acquisitions using appropriated funds, including services and new requirements. FAR 23.404(b) applies and states the GPP requires 100% of EPA designated product purchase that are included in the Comprehensive Procurement Guidelines list that contains recovered materials, unless the item cannot be acquired: a) competitively within a reasonable timeframe; b) meet appropriate performance standards, or c) at a reasonable price. The prime contractor is responsible for ensuring that all subcontractors comply with this requirement.

5. PRODUCT CONFORMANCE PROVISIONS.

5.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First production inspection (see 5.2).

b. Conformance inspection (see 5.3).

5.2 First production inspection. The first production vehicle shall be subjected to the examinations and tests described in 5.6.3.1 through 5.6.3.5 (if applicable). The contractor shall provide or arrange for all test equipment, personnel, and facilities.

5.3 Conformance inspection. Each production vehicle shall be subjected to the examinations and tests described in 5.6.3.1, 5.6.3.1.5, 5.6.3.1.8, 5.6.3.1.9, 5.6.3.1.11, 5.6.3.1.3, 5.6.3.1.12, and 5.6.3.3.1 through 5.6.3.5 (if applicable).

5.4 Product conformance. The products provided shall meet the salient characteristics of this CID, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial marketplace. The government reserves the right to require proof of such conformance.

5.5 Commercial item requirement. The vehicle furnished shall comply with the "commercial item" definition of FAR 2.101 as of the date of award. The Government reserves the right to require the manufacturer to prove that their product complies with the referenced commerciality requirements and each salient characteristic of this CID. The manufacturer shall provide an itemized technical proposal that describes how the proposed model complies with each salient characteristic of this CID**;** a paragraph by paragraph response to the salient characteristics section of this CID shall be provided**.** The response to each paragraph shall clearly state “Fully Comply” or “Bid with Exception” in addition to other data the contractor wishes to submit. The proposal shall be provided with the pricing submission. Failure to provide this information may deem a vendor as non-responsive and their proposal may be rejected. The manufacturer shall provide two copies of their commercial descriptive catalogs with their offer as supporting reference to the itemized technical proposal. The manufacturer shall identify all modifications made to their commercial model in order to comply with the requirements herein.

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5.6 Inspection requirements.

5.6.1 General inspection requirements. Apparatus used in conjunction with the inspections specified herein shall be laboratory precision type, calibrated at proper intervals to ensure laboratory accuracy.

5.6.2 Test rejection criteria. Throughout all tests specified herein, the vehicle shall be closely observed for the following conditions, which shall be cause for rejection:

a. Failure to conform to design or performance requirements specified herein or in the contractor's technical proposal.

b. Any spillage or leakage of any liquid, including fuel, coolant, lubricant, or hydraulic fluid, under any condition, except as allowed herein.

c. Structural failure of any component, including permanent deformation, or evidence of impending failure.

d. Evidence of excessive wear.

e. Interference between the vehicle components or between the vehicle, the ground, and all required obstacles, with the exception of normal contact by the tires.

f. Misalignment of components.

g. Evidence of undesirable roadability characteristics, including instability in handling during cornering, braking, and while traversing all required terrain.

h. Conditions that present a safety hazard to personnel during operation, servicing, or maintenance.

i. Overheating of the engine, transmission, or any other vehicle component.

j. Evidence of corrosion.

k. Failure of the firefighting system.

5.6.3 Detailed inspection requirements.

5.6.3.1 Examination of product. Each vehicle shall be examined to verify compliance with the requirements herein. A Government generated checklist that identifies each relevant requirement and the inspection results shall be used. Particular attention shall be given to materials, workmanship, dimensions, surface finishes, protective coatings and sealants and their application, welding, fastening, and markings. Proper operation of each vehicle function shall be verified. A copy of the vehicle manufacturer's certifications shall be provided with each vehicle

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in accordance with NFPA 414. Each production vehicle shall be inspected to a reduced version of the checklist in accordance with NFPA 414. The cost associated with third party independent certification testing is the responsibility of the vehicle manufacturer.

5.6.3.1.1 Roadability test. The fully loaded first production vehicle shall be driven over ten miles of paved and ten miles of cross country terrain consisting of open fields, broken ground, and uneven terrain to demonstrate compliance with 3.1.6.1. After completing the driving portion of the roadability test, all loads shall be removed and all structure and surfaces shall be visibly inspected for failure or permanent deformation.

5.6.3.1.2 Gradeability test. The fully loaded first production vehicle shall be tested in accordance with NFPA 414 to demonstrate compliance with 3.1.6.2.

5.6.3.1.3 Tilt table test. The fully loaded first production vehicle shall be tested in accordance with NFPA 414 to demonstrate compliance with 3.1.6.3. A slip/trip rail with a maximum height of 2-inches may be used. If an adjustable height suspension system is provided, the suspension system may be set to the height normally used on hard pavement.

5.6.3.1.4 Cornering stability test. The fully loaded first production vehicle shall be tested in accordance with NFPA 414 to demonstrate compliance with 3.1.6.4.

5.6.3.1.5 Weight and weight distribution measurement. Each vehicle shall be weighed (full and empty) to demonstrate compliance with NFPA 414. A certification shall be provided.

5.6.3.1.6 Dimension measurement. The first production vehicle shall be measured in accordance with NFPA 414. In addition to the dimensions listed in NFPA 414, the overall length, width, and height shall be measured to demonstrate compliance with 3.2.1.

5.6.3.1.7 Angles of approach and departure measurement. The angles of approach and departure of the fully loaded vehicle shall be measured to demonstrate compliance with 3.2.2.

5.6.3.1.8 Acceleration test. Each vehicle shall be tested in accordance with NFPA 414 to demonstrate compliance with 3.3.1.2. For the first production vehicle, a manufacturer supplied time-distance recorder shall be used to record data for this test.

5.6.3.1.9 Maximum speed test. Each vehicle shall be tested in accordance with 5.4.4 of NFPA 414 to demonstrate compliance with 3.3.1.3. For the fully loaded first production vehicle, a time-distance recorder shall be used to record data for this test.

5.6.3.1.10 Pump and roll on a 40-percent grade test. The fully loaded first production vehicle shall be tested in accordance with NFPA 414.

5.6.3.1.11 Service brake system test. Each vehicle shall be tested in accordance with NFPA 414. For the fully loaded first production vehicle, a time-distance recorder shall be used to record data for this test.

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5.6.3.1.12 Air system and air compressor test. Each vehicle shall be tested in accordance with NFPA 414.

5.6.3.1.13 Turning diameter test. The fully loaded first production vehicle shall be tested in accordance with NFPA 414 to demonstrate compliance with 3.3.12.2.

5.6.3.2 Cab interior sound level test. The cab interior sound levels of the first production vehicle shall be measured in accordance with NFPA 414 to demonstrate compliance with 3.4.2.

5.6.3.3 Agent system tests.

5.6.3.3.1 Agent discharge pumping test. Each vehicle shall be tested in accordance with NFPA 414.

5.6.3.3.2 Pump and maneuver test. Each vehicle shall be tested in accordance with NFPA 414.

5.6.3.3.3 Hydrostatic pressure test. Each vehicle shall be tested in accordance with NFPA 414.

5.6.3.3.4 Priming device test. Each vehicle that is equipped with a priming pump shall be tested in accordance with NFPA 1901.

5.6.3.3.5 Agent tank capacity test. The first production vehicle shall be tested in accordance with NFPA 414 to demonstrate compliance with the capacity requirements of 3.6.2 and 3.6.3.1.

5.6.3.3.6 Water tank fill and overflow test. The first production vehicle shall be tested in accordance with NFPA 414 to demonstrate compliance with 3.6.2.3.

5.6.3.3.7 Foam concentration test. Each vehicle shall be tested in accordance with NFPA 414 at the 3-percent concentration setting to demonstrate compliance with the foam concentration quality requirements of 3.6.4.

5.6.3.3.8 Primary turret flow rate test. The first production vehicle shall be tested in accordance with NFPA 414. Each vehicle shall be tested to demonstrate compliance with NFPA 414.

5.6.3.4 Electrical system tests.

5.6.3.4.1 Electrical charging system test. The first production vehicle shall be tested in accordance with NFPA 414 to demonstrate compliance with Annex B.4 of NFPA 414.

5.6.3.4.2 Electromagnetic interference test. The first production vehicle shall be tested in accordance with NFPA 414, IEC CISPR 12, and IEC CISPR 25 to demonstrate compliance with 3.8.4.

5.6.3.5 Piercing/penetrating nozzle test. Each truck equipped with a piercing/penetrating nozzle shall be tested in accordance with NFPA 414.

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5.6.3.6 Ancillary Equipment Inventory. The first production vehicle will have a detailed ancillary equipment inventory list. All items will be inventoried and repackaged per 3.5.5.

6. PACKAGING.

6.1 Preservation, packing, and marking shall be as specified in the contract or order.

7. NOTES.

7.1 Source of documents.

7.1.1 Department of Defense and Federal documents may be obtained at https://assist.daps.dla.mil or from the Document Automation and Production Service, Bldg 4D (DPM-DODSSP), 700 Robbins Avenue, Philadelphia PA 19111-5094.

7.1.2 The Code of Federal Regulations (CFR) may be obtained at http://www.gpoaccess.gov/cfr/ or from the Superintendent of Documents, U.S. Government Printing Office, Washington DC 20402.

7.1.3 SAE documents may be obtained http://www.sae.org/servlets/index or from SAE, Inc., 400 Commonwealth Drive, Warrendale PA 15096.

7.1.4 NFPA documents may be obtained at http://www.nfpa.org/index.asp or from NFPA, Batterymarch Park, Quincy MA 02269-9101.

7.1.5 International Electrotechnical Commission (IEC) standards may be obtained from IEC Central Office, 3, rue de Varembé, P.O. Box 131, CH - 1211 GENEVA 20, Switzerland. Electronic copies of IEC standards may be obtained from http://www.iec.ch/

7.1.6 FAR and DFARS may be obtained from the Superintendent of Documents, P.O. Box 371954, Pittsburgh, PA 15250-7954. Electronic copies of the FAR may be obtained from http://www.arnet.gov/far Electronic copies of the DFARS may be obtained from http://www.acq.osd.mil/dpap/dars/dfars/index.htm .

7.1.7 Occupational Safety and Health Association (OSHA) copies may be obtained online at http://www.osha.gov/ or from OSHA, 200 Constitution Ave., Washington, D.C. 20210.

7.1.8 Environmental Protection Agency documents may be obtained at http://www.epa.gov/ or from USEPA Ariel Rios Building (AR), 1200 Pennsylvania Avenue N.W., Washington, DC 20004.

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7.2 Ordering data. The contract or order should specify the following:

a. Finish color required (Desert Sand in place of Candy Apple Red) (see 3.1.2.1).

b. If a standard roof turret or an extendable turret is required (see 3.6.5). (NOTE: The option for an extendable turret is only applicable at NFPA Category 9 and 10 locations. Bases that fall into a lower category are not authorized the extendable turret option). If an extendable turret is selected it will be provided with the following standard items
(1) A mounted video camera and cab-mounted monitor (see 3.6.5.2.1).
(2) A mounted aircraft skin penetrator (see 3.6.5.2.2).
(3) A dry chemical primary turret discharge nozzle (see 3.7.1).

c. Emergency warning light color required (amber or blue in place of red) (see 3.8.7.1). (NOTE: Applies to USAFE only)

d. Electrical shoreline voltage, 220 volts instead of 110 volts (see 3.9.1). (NOTE: Applies to USAFE only)

e. Engine fuel type (diesel or ultra low sulfur diesel) and units of volume (gallons or liters)

7.3 Key Words.

Gallon per minute
Rapid intervention
Ultra High Pressure

Custodian:
Air Force - 84

Reviewer:
DLA-CC

Preparing activity:
Air Force - 84

Agent:
Air Force – 99

Project 4210-2010-011

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.daps.dla.mil .

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